

KPP-01 Buoyancy-Driven Pneumatic DEMO SIMULATOR - Governing Physics

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Document type: Scientist-grade physics note for the live HMI/simulator
Plant / code: D:\PNEUMATIC project v1 (hmi_demo)
Live HMI: <http://127.0.0.1:4173/>
Grounding snapshot: GET <http://127.0.0.1:4173/api/grok/snapshot>
capturedAt: 2026-09-18T19:23:45.567Z
Fidelity: engineering | demoAssist live: 0 N|m | Plant status: running

> Grounding file (Dina): C:\Users\Nizar\AppData\Local\Temp\kpp_snap_out.txt - outer capturedAt 2026-09-18T19:23:45.568Z, snapshot 2026-09-18T19:23:45.567Z (RUNNING, engineering, receiverSP=8, injReq=1.9, demoAssist=0).

> Every numeric claim below is taken from that snapshot's live tags and/or designBasis / designBasisAdvice / designBasisWarnings. No invented plant constants.

Abstract

KPP-01's HMI simulator integrates open-cycle pneumatic injection, hydrostatic floater gas state, buoyant torque on an endless chain, rotational rigid-body dynamics, geared generation, and an energy ledger each tick. Design-basis net plant power is 13.257 kW (compressorPowerKW = 15.843, designElectricalKW = 2.586). The plant is air-supply limited near 2.59 kW electrical - not the 8 kW generator nameplate. With receiver cut-out 8 bara and dock injection need 1.90 bara, about 11.0 kW is throttled at the regulator (~69% of design compressor shaft power). In engineering fidelity, demoAssist == 0 and must never be labelled buoyancy.

Keywords: buoyancy; open-cycle pneumatics; Boyle/polytropic floater gas; hydrostatic injection; throttle loss; energy balance; KPP-01 simulator

0. Grounding snapshot (authoritative)

Source file: C:\Users\Nizar\AppData\Local\Temp\kpp_snap_out.txt
snapshot.capturedAt = 2026-09-18T19:23:45.567Z | status=running | fidelity=engineering

Symbol / tag	Value	Role in equations	
P_{rec} setpoints.receiverBara	8 bara	compressor cut-out	
P_{reg} setpoints.regulatorBara	2.3 bara	header setpoint	
P_{dock} injection.requiredBara / designBasis.injectionPressureBara	1.9 / 1.9 bara	hydrostatic fill head	
\tau_b torqueNm.buoyancy	7856.1 N	m	buoyant drive
\tau_{\mathrm{assist}} torqueNm.demoAssist	0 N	m	must stay 0 in eng.
n_{\mathrm{tw}} towerRPM	2.755 rpm	\omega=2\pi n/60	
P_{\mathrm{gen}} deliveredKW	1.817 kW	live electrical	
P_{\mathrm{gen,des}} designElectricalKW	2.586 kW	air-limited design	
P_{\mathrm{comp,des}} compressorPowerKW	15.843 kW	design compression	
P_{\mathrm{net,des}} netPlantPowerKW	-13.257 kW	P_{gen}-P_{comp}	
nameplate ratedKW	8 kW	not delivered	
gear liveRatio	108.9	\omega_g \approx i\omega	
air mass injected/vented	127.81 / 126.237 kg	open-cycle balance	

Throttle fraction from designBasis warning (8 bar vs 1.90 bara):

$$\frac{15.843-4.89}{15.843} \approx 0.691 \text{ \ (69\%)}$$

1. What the plant is

KPP-01 is an open-cycle buoyancy pneumatic demonstration:

- 1. Compressor fills a receiver.
- 2. Regulator drops pressure to a header setpoint.
- 3. At the dock (bottom), air is injected into floaters against hydrostatic head.
- 4. Air-filled floaters ascend; empty/vented floaters descend.
- 5. Net buoyant torque turns the tower -> gearbox -> generator.
- 6. At the top, air is vented to atmosphere (open cycle). Compression work in that air is largely rejected.

Simulator modules (code): hmi_demo/js/utlis/FloaterPhysics.js, hmi_demo/js/models/TowerDynamics.js, hmi_demo/js/utlis/Config.js (pressureAtDepth, designBasis builders), hmi_demo/js/utlis/PlantSnapshot.js (API exposure).

2. Nomenclature

Symbol	Meaning	SI / plant unit
----- ----- -----		
P_{atm}	Surface absolute pressure	bara
h	Depth below free surface	m
ρ	Water density (WATER_DENSITY)	1000 kg/m ³
g	Gravity	m/s ²
P_{dock}	Absolute injection pressure at dock	bara
V	Displaced / gas volume of a floater	m ³
m	Floater structure mass (plus entrained terms in design)	kg
F_b	Net buoyant force	N
r	Sprocket radius (sprocketRadiusM)	m
τ	Torque	N m
ω	Angular velocity	rad/s
I	Tower rotational inertia	kg m ²
$\dot{V}_{\mathrm{FAD}}$	Compressor free-air delivery	m ³ /min
P_{gen} , P_{comp} , P_{net}	Electrical out, compressor in, net	kW
τ_{assist}	demoAssist torque (not buoyancy)	N m

3. Hydrostatics at the dock

3.1 Governing equation

Fresh-water hydrostatic gradient in the simulator is 0.1 bar/m (BAR_PER_METRE = 0.1 in FloaterPhysics.js):

$$P(h) = P_{\mathrm{atm}} + (0.1 \, \mathrm{bar/m}) h$$

equivalently

$$P_{\mathrm{dock}} = P_{\mathrm{atm}} + \rho g h$$

$$\frac{\rho g}{10^5} \approx 0.0981 \, \mathrm{bar/m} \approx 0.1 \, \mathrm{bar/m}.$$

Code: ambientPressureAt(depthM, atmBara) and pressureAtDepth(depthM) in Config.js.

3.2 Snapshot / designBasis numbers

Quantity	Value	Source
----- ----- -----		
designBasis.injectionPressureBara	1.9	designBasis
Live pneumatic.injection.requiredBara (RUNNING)	1.9	(no active deep fill demand this frame) live
Implied dock depth at 1.9 bara if $P_{\mathrm{atm}}=1$	$h=(1.9-1)/0.1=9$	m derived

Physics truth for running fill: designBasis injection head is 1.90 bara.

4. Boyle / polytropic free-air volume for floater fill

4.1 Isothermal (Boyle) mass-volume relation

Default gas law in FloaterPhysics.js is isothermal:

$$m = \rho_1 V$$

$$\quad \rightarrow \quad V = \frac{m}{\rho_1 P}$$

with P in bara and ρ_1 = air density at 1 bara.

Free-air equivalent volume to fill geometric volume V_f at dock pressure:

$$V_{\mathrm{free}} = V_f \cdot \frac{P_{\mathrm{dock}}}{P_{\mathrm{atm}}}$$

Example: if $V_f = 0.16 \, \mathrm{m}^3$ and $P_{\mathrm{dock}} = 1.9 \, \mathrm{bara}$, $V_{\mathrm{free}} = 0.304 \, \mathrm{m}^3$ per fill (illustrative geometry; live inventory uses mass tags).

4.2 Polytropic extension

For exponent $n \in [1, 1.4]$:

$$V = V_{\mathrm{ref}} \left(\frac{P_{\mathrm{ref}}}{P} \right)^{1/n}$$

$(n=1)$ recovers Boyle; $(n>1)$ reduces mean ascending volume (less buoyancy work per injected mass). Code: `expansionVolume(...)`.

4.3 Live air inventory (this snapshot)

Tag	Value
accounting.injected/vented	127.81 / 126.237 kg
pneumatic.accounting.injectedKg	2.156 kg
ventedKg / recoveredKg	0 / 0 (plant ready; no active lap vent this frame)

5. Buoyant force, torque, and shaft power

5.1 Force on one floater

$$F_b = (\rho V - m)g$$

Ascending (air volume V): typically $(F_b > 0)$ (up).

Descending (vented / water-filled): buoyancy shrinks; weight dominates.

Design builder in Config.js uses the same idea for mean ascending force:

$$F_{\mathrm{asc,mean}} = \bar{V}_{\mathrm{air}} \rho g - W_{\mathrm{structure}}$$

5.2 Torque sum on the sprocket

$$\tau_{\mathrm{buoyancy}} = \sum_i F_{b,i} r_i$$

$$(r \approx \text{sprocketRadiusM})$$

Critical fidelity rule:

$$\tau_{\mathrm{driving}} = \tau_{\mathrm{buoyancy}} + \tau_{\mathrm{assist}}$$

with $(\tau_{\mathrm{assist}} = \text{demoAssist})$ - not buoyancy.

Quantity	Value	Source
Live torqueNm.buoyancy	7856.1 N m	snapshot
Live torqueNm.demoAssist	0 N m	snapshot
designBasis.designTorqueNm	10796.809 N m	designBasis
designBasis.demoAssistTorqueNm	17841.538 N m	what-if only - not live eng torque

5.3 Mechanical power

$$P_{\mathrm{mech}} = \tau \omega, \quad \omega = 2\pi n/60$$

Design check:

$$P_{\mathrm{mech,design}} = 10796.809 \times \frac{2\pi \times 2.755}{60} \approx 3.114 \text{ kW}$$

matches `designBasis.designMechanicalKW = 3.114`.

Live (ready/braked): `towerRPM = 0`, so $(P_{\mathrm{mech,live}} = 0)$ this frame; buoyancy torque is held against brake (brake = 14500 N|m).

5.4 Rotational dynamics each tick (TowerDynamics.updateDynamics)

Newton's second law for rotation:

$$I \alpha = \tau_{\mathrm{net}}, \quad \alpha = \frac{\tau_{\mathrm{net}}}{I}$$

$$\omega \rightarrow \omega + \alpha, \Delta t$$

Resistive torques (friction, water drag, brake, generator load) oppose motion; they stop the shaft but do not reverse-drive it. Code zero-cross clamps prevent chatter.

Chain speed:

$$v_{\text{chain}} = \omega \cdot r_{\text{sprocket}}$$

6. Pneumatic chain: receiver, regulator, FAD vs demand

6.1 Snapshot setpoints & sensors

Quantity	Value	Source
setpoints.receiverBar	8	live
setpoints.regulatorBar	2.3	live
sensors.ptTankBar	6.936	live
sensors.ptHeaderBar	2.095	live
Compressor FAD	3.2 m³/min	live + designBasis
designBasis.airDemandM3Min	3.2	designBasis
designBasis.compressorPowerKW	15.843	designBasis

6.2 Why 8 bara vs ~1.9 dock wastes ~69%

Idealized isothermal compressor work to pressure P scales as $P \ln(P/P_0)$ (or catalogue curve at cut-out). DesignBasis warning states explicitly:

> Receiver set to 8 bar but injection only needs 1.90 bara: 15.84 kW of compression where 4.89 kW would do. The surplus is throttled away at the regulator.

$$\eta_{\text{throttle, waste}} = \frac{15.843 - 4.89}{15.843} \approx 0.691 \text{ (69\%)}$$

designBasisAdvice: ?About 11.0 kW is throttled away at the regulator.? Suggestion: cut-out 2.3 bar (dock + 0.4 bar).

Regulator drop:

$$\Delta P_{\text{throttle}} = P_{\text{receiver}} - P_{\text{header}}$$

is irreversible for plant electrical output: shaft power spent raising pressure above dock need does not increase buoyant V at the dock.

7. Open-cycle vent mass and energy

7.1 Mass balance tags

$$m_{\text{injected}} = m_{\text{vented}} + m_{\text{held}} + m_{\text{spilled}} - m_{\text{recovered}} + \epsilon$$

This snapshot (ready): injected 2.156 kg, vented 0, recovered 0, balanceErrorKg = 0.

During a running lap, design/intent is vent at top -> open cycle (recoveredKg remains 0 unless a recovery model is enabled).

7.2 Energy ledger (energyKWh)

Channel	kWh (this snapshot window)
in	0.0753
out	0
compressionLoss	0.0226
throttle	0
spillVent	0.0022
assist	0
unexplainedFrac	0.5178 (short/idle window - not a steady run)

On a long engineering run, throttle and compression dominate in ? out, consistent with designBasis.

8. Generator path: air-limited ~2.6 kW vs 8 kW nameplate

```
| Quantity | Value | Source |
|-----|-----|-----|
| electrical.generator.ratedKW | 8 | live |
| designBasis.designElectricalKW | 2.586 | designBasis |
| electrical.generator.loadRequestKW | 2.6 | live |
| Live delivered (RUNNING) | 1.817 kW | tower stopped |
```

Advice: ?An 8 kW generator at this torque needs about 9.90 m³/min FAD? selected compressor delivers 3.20 m³/min, which supports 2.59 kW.?

```
\[
P_{\mathrm{elec,air\text{-}limit}} \approx 2.59\,\mathrm{kW} \parallel 8\,\mathrm{kW}_{\mathrm{nameplate}}
\]
```

Gear: fittedGearRatio = 108.9 (design recommended 108.911) - matches gen ~300 rpm at tower ~2.755 rpm when running.

9. Net plant power

```
\[
P_{\mathrm{net}} = P_{\mathrm{gen}} - P_{\mathrm{comp}}
\]
```

```
| Basis | \((P_{\mathrm{gen}}) | \((P_{\mathrm{comp}}) | \((P_{\mathrm{net}}) |
|-----|-----|-----|-----|
| designBasis | 2.586 kW | 15.843 kW | ?13.257 kW |
| live plantPower (this frame; compressor coasting) | 1.817 | 0 kW | 1.817 kW |
```

designBasisWarnings: ?Design-basis net plant power is -13.26 kW (2.59 kW generated vs 15.84 kW of compression).?

Interpretation: negative net is expected open-cycle physics with over-pressured receiver - not a sensor bug.

10. demoAssist vs buoyancy (engineering fidelity)

```
| | Live engineering | designBasis field | |
|---|---|---|---|
| Buoyancy torque | 10779.7 N|m | designTorqueNm 10796.809 |
| demoAssist | 0 N|m | demoAssistTorqueNm 17841.538 (counterfactual) |
| energyKWh.assist | 0 | - |
| fidelityMode | engineering | - |
```

Rule: Never call demoAssist buoyancy. Engineering acceptance requires $\tau_{\mathrm{assist}} \equiv 0$.

11. What the simulator is solving each tick

State vector (conceptual; tags in State.js):

```
\[
\mathbf{x} = \big\{
\omega,\,\theta,\,
\{m_i,V_i,\text{segment}_i\}_{i=1}^N,\,
P_{\mathrm{tank}},P_{\mathrm{header}},\,
\text{valve/compressor flags},\,
E_{\mathrm{ledger}},\,
\text{gen load}
\big\}
\]
```

One (Δt) step (engineering fidelity):

1. Hydrostatics / gas state - for each floater, $(P_{\mathrm{amb}})(h)=P_{\mathrm{atm}}+0.1(h)$; update mass/volume via Boyle/polytropic (FloaterPhysics).
2. Dock injection - if docked and header above requiredBara, add air mass; demand accumulates toward FAD limit.
3. Vent / spill - at top segment, open-cycle mass leave; accounting tags update.
4. Forces -> torques - $(F_{b,i})=(\rho V_i-m_i)g$; $(\tau_{\mathrm{buoy}})=\sum F_{b,i} r_i$; add demoAssist only if enabled (0 in eng.).
5. Resistive torques - friction, bearing, water drag, brake, generator load.
6. Rigid-body update - $(\alpha)=\tau_{\mathrm{net}}/I$; integrate (ω,θ) ; derive RPM & chain speed (TowerDynamics.updateDynamics).
7. Pneumatic plant - compressor toward receiver setpoint; regulator toward header setpoint; throttle loss enters energy ledger.

8. Electrical - gearbox ratio maps $\omega_{\text{to}} \omega_{\text{g}}$; delivered kW $\leq$ air-limited / torque-limited available power; never silently equal to 8 kW nameplate.
9. Energy ledger - integrate compressor in, gen out, compression/throttle/vent/drag/friction/brake/gearbox/converter/assist channels.
10. Snapshot API - PlantSnapshot publishes tags + recomputed designBasis block for GET `/api/grok/snapshot`.

Loop geometry this basis: loopLengthM = 21.705, floaterPitchM = 0.748, N = 29 floaters, lapTimeSec = 86.989 at design RPM, fillsPerMin = 20.003.

12. Equation index (quick)

1. $P_{\text{dock}} = P_{\text{atm}} + \rho g h$ (0.1 bar/m in code)
2. $V_{\text{free}} = V_f P_{\text{dock}} / P_{\text{atm}}$ (Boyle fill)
3. $F_b = (\rho V_m g)$; $\tau = \sum F_b r$; $P = \tau \omega$
4. $\alpha = \tau_{\text{net}}$; $\omega \rightarrow \omega + \alpha \Delta t$
5. Throttle waste $\approx (15.843 - 4.89) / 15.843 \approx 69\%$ at receiver 8 vs dock 1.9
6. $P_{\text{net}} = P_{\text{gen}} - P_{\text{comp}}$ -> designBasis ?13.257 kW
7. $\tau_{\text{assist}} \equiv 0$ in engineering (!= buoyancy)

13. Conclusions

1. The simulator is a coupled hydrostatic-pneumatic-mechanical-electrical integrator, not a cartoon.
2. Design-basis net ?13.257 kW with air-limited electrical 2.586 kW is the honest stock plant.
3. ~69% of design compressor power is regulator throttle from receiver 8 bara vs dock 1.9 bara.
4. 8 kW is nameplate ceiling, not delivered physics.
5. demoAssist = 0 in engineering; designBasis assist torque is a what-if only.
6. Net ≥ 0 requires pressure matching and vent recovery / closed cycle - not a bigger generator sticker.

14. Data citation

- Snapshot: capturedAt = 2026-09-18T19:23:45.567Z via GET `http://127.0.0.1:4173/api/grok/snapshot`
- Blocks used: plant, pneumatic, mechanical.torqueNm, electrical, energyKWh, designBasis, designBasisAdvice, designBasisWarnings
- Code references: `hmi_demo/js/utis/FloaterPhysics.js`, `Config.js` (pressureAtDepth, WATER_DENSITY), `models/TowerDynamics.js` (updateDynamics)
- Author actions: read-only; no plant START/STOP/setpoint writes for this document.
- End -